Zhu Chunzheng

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Gender: Male

Age: 24

education experience

2018.9–2022.6 **Central University For Nationalities**

Data Science and Big Data Technology | Bachelor's Degree

2022.9–2024.6 **Hunan University**

Computer Science and Technology | Master's Degree | Research Focus: Image Quality Assessment, Brain-Computer Interface

September 2024 – Present **Hunan University**

Computer Science and Technology | Ph.D. | Research Focus: AI-Assisted Fetal Ultrasound Diagnosis and MLLMs

Competitive and Research Experience

Competitions: Second Prize in the 2019 Beijing University Student Animation Design Competition; Second-Class Academic Scholarship at Minzu University of China (2020–2021); First-Class Academic Scholarship at Minzu University of China (2021–2022); M Prize in the 2020 Mathematical Contest in Modeling (MCM); Second Prize in the Higher Education Press Cup " National Undergraduate Mathematical Modeling Competition" (Beijing); Second Prize in the 2020 China Computer Design Competition (Beijing); Second Prize in the Graduate Group of the 2023 Mathor Cup University Mathematical Modeling Challenge Big Data Competition; Third Prize in the 20th China Graduate Mathematical Modeling Competition; First Prize in the 19th Challenge Cup "AI+" Special Competition – a globally leading AI+ ultrasound technology solution.

Admitted papers:

1、 Zhu C., Li NY., Chen S., Lin J., Wang Y. Med Eyes: Learning Dynamic Visual Focus for Medical Progressive Diagnosis. AAAI Conference on Artificial Intelligence, AAAI, 2026. CCF-A Category

Introduction: To address the low utilization rate of visual information in medical diagnostics, this study proposes a framework integrating multimodal large models with reinforcement learning. By simulating the dynamic visual focusing process of physicians during "scan-drill" procedures, the dual-stream GRPO framework significantly enhances the accuracy and interpretability of medical visual question answering (VQA).

2 、 Zhu C., Li NY., Shao J., Lin J. Pathology-Aware Prototype Evolution via LLM-Driven Semantic Disambiguation for Multicenter Diabetic Retinopathy Diagnosis, ACM MM, 2025. CCF Class A

Introduction: To address the long-tail distribution and domain-specific variations in multi-center retinal disease medical data, this study employs large language models (LLMs)to generate fine-grained pathological semantic descriptions, which assist in a two-stage visual prototype evolution process. This approach effectively resolves the challenges of identifying subtle features and achieving cross-domain generalization in diabetic retinopathy grading.

3 、 Zhu C., Shao J., Lin J., Wang Y., Wang J., Tang J., Li K. fMRI2GES: Co-speech Gesture Reconstruction from fMRI Signals with Dual Brain Decoding Alignment, TCSVT, 2025. Area 1 Top

Introduction: This study pioneered the exploration of reconstructing co-speech gestures from fMRI brain signals, proposing a dual-brain decoding alignment mechanism and a conditional diffusion model to achieve unsupervised cross-modal alignment without paired data.

4、 Zhu C., Lin J., Tan G., Zhu N., Li K., Wang C., Li S. Advancing Ultrasound Medical Continuous Learning with Task-Specific Generalization and Adaptability, BIBM, 2024. CCF-B Category

Introduction: To address the problem of catastrophic forgetting in ultrasound medical image analysis, a continuous learning framework based on Masked Modeling and knowledge distillation has been developed. This framework achieves efficient adaptation to new disease types while maintaining performance on existing tasks.

5、Lin J.*, Zhu C.*, Kneuertz P., Bai Y., Xue Y. When Models Learn to Ask Why: Adaptive Causal Reasoning for Trustworthy Medical Vision – Language Models, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR Findings 2026), CCF-A Category

Introduction: To address the hallucination issues in large medical models, an adaptive causal reasoning mechanism is introduced. By utilizing the reconstructed structured causal chain dataset CRMed and employing a two-stage SFT+RL optimization approach, the multimodal model enables active "causal analysis and verification," thereby enhancing diagnostic credibility and robustness.

6、Zhu C., Wang Y., Lin J.*, Wang F., Wang H., Zhao L., Li S., Li K., Anatomy-Anchored Self Supervision: Distilling Vision Foundation Models for Invariant Ultrasound Representation, (MICCAI), 2026.

Introduction: This study introduces AnaUS, the first anatomical-level self-supervised pre-training framework, which achieves unlabeled structural discovery through a learning-based latent prompt engine-driven LP-SAM, combined with dual-strategy self-supervised learning (anatomical separation alignment and core region prediction), delivering SOTA performance across six public ultrasound benchmarks.

7、Zhu C., Lin J.*, Wang F., Jiang C., Tan G., Zhou Z., Li S., Li K., Subspace-Guided Semantic and Topological Invariant Registration for Annotation-Free Ultrasound Plane Quality Control, (MICCAI), 2026. CCF-B Category

Introduction: STRIQ is proposed—a registration-driven, label-free ultrasound plane quality control framework that achieves optimal correlation with clinical quality scores on the US4QA and CAMUS datasets through its Potential Registration Aligner (LRA) and Orthogonal Knowledge Subspace (OKS) modules.

Project Experience

September 2023 – Present **Artificial Intelligence-Focused Fetal Ultrasound-Assisted Diagnosis**

Developed by the Ultrasound Research Team at Hunan University

The engineering and research content includes: large-scale classification models, disease detection, self-supervised learning, continuous learning, and development of medical large-scale models.

Educational Background

2019.09–2022.06	League Secretary	**Central Minzu University Data Science and Big Data Technology Program**
2020 09–2021 06	President of the Badminton Association	**Central Minzu University Badminton Association**

certificate of honor

Outstanding Member of the Communist Youth League at Minzu University of China in 2020; Outstanding Graduate Student at Hunan University for the academic years 2023–2024 and 2024–2025.